Nexus Learning Management System (LMS)

Complete Technical Feature Specification & Postman API Reference Guide

Version: 3.2.0 (Enterprise Live)	**Stack:** FastAPI (Async Python) + SQLite/Postgres	**Public Cloudflare:** Live Tunnel
Local Server: http://127.0.0.1:8000	**API Standard:** OpenAPI 3.0 / Postman v2.1	**Repository:** github.com/rishrivastava/nexus-lms

1. Executive Architectural Overview

Nexus LMS is an ultra-performant, single-binary Python ASGI learning platform tailored for competitive exams, coaching academies, K-12 networks, and higher education. Unlike legacy PHP or distributed microservices architectures, Nexus delivers sub-50ms page load speeds, native Generative AI capabilities (Google Gemini), subjective handwritten PDF answer sheet evaluations, and strict departmental faculty governance.

2. Strategic Comparison: Nexus LMS vs. Moodle vs. Open edX

The following matrix summarizes the architectural, operational, and pedagogical distinctions between Nexus LMS and open-source alternatives:

Dimension / Feature	Nexus LMS (Our Platform)	Moodle (Open Source)	Open edX (Open Source)
Architecture	FastAPI (Async Python) + SQLite/Postgres + TailwindCSS	Monolithic PHP + Apache/MySQL (LAMP Stack)	Distributed Django + Studio + K8s / Tutor Microservices
Resource Footprint	**512MB – 1GB RAM** ($4/mo VPS), <50ms response	**2GB – 4GB RAM**, high DB query overhead	**16GB – 32GB RAM**, heavy idle multi-container load
Ans Sheet Evaluation	**Built-in first-class**: Handwritten PDF uploads, faculty marks, model answers, and Admin Surveillance Window.	Generic assignment module; requires PDF annotation plugin.	Lacks native subjective handwritten paper workflows (code/MCQ only).
Faculty Governance	**Strict Hierarchy**: Teacher assigned to departments first → only department courses assignable → strict class isolation.	Flat role assignment; loose course enrollments without dept locking.	Studio role controls; lacks dynamic multi-department academic buckets.
Native AI Engine	**Built-in Gemini AI**: Dynamic Blooms-taxonomy quiz generation, contextual doubt solver.	None native; relies on commercial third-party plugins.	None native in core distribution; experimental external LTI.
Live Classes & Doubts	**Native module**: WebRTC/Meet integration, in-stream doubt queue, live attendance tracking.	Requires external BigBlueButton or Zoom server cluster.	Requires external Zoom / Teams LTI microservices.

1-on-1 Mentorship	**Native booking engine**: Student appointment slots, calendar sync, counseling notes.	Requires separate scheduler activity plugins.	Not supported natively (strictly 1-to-many MOOC).
National K-12 Schema	**Built-in**: UDISE codes, School Block, District, State, Guardian records.	Requires manual custom profile field expansion.	Minimal user model (email, country); no K-12 school hierarchy.

3. Core Technical & Operational Innovations

3.1 Subjective Ans Sheet Upload & Administrative Surveillance Window

Competitive examinations and school boards require subjective handwritten evaluation. Nexus LMS incorporates a zero-plugin subjective engine: students photograph or scan handwritten answer copies as PDFs. When uploaded, the PDF instantly appears on the course teacher's profile under *'Enrolled Student Answer Sheets & Evaluations'*. Simultaneously, the Department Administration accesses a dedicated **'Answer Sheets Monitor'** tab with 4 live surveillance metrics (Total Received, Evaluated Copies, Pending Evaluation, and Evaluation Rate %) and a filterable grading dashboard.

3.2 Hierarchical Faculty Governance & Department Course Alignment

To ensure institutional compliance and academic isolation, faculty course assignments follow a strict flow: **(1) Multi-Department Assignment**: Admin assigns a faculty member to one or more department buckets (e.g. Test Prep, Foreign Language). **(2) Department-Filtered Course Assignment**: Admin can only assign courses belonging to the teacher's active departments. Course creation/editing restricts reassignment options to valid department faculty. **(3) Strict Class Isolation**: Teachers strictly view only their explicitly assigned classes across Teacher Studio, Analytics, and Profile.

3.3 Clean User Name Addressal

All user headings and greetings across portals strictly address users by full name (e.g. Welcome Priya Patel!, Welcome Prof. Rameez!, Welcome Rishu!) without redundant role prefixes like 'Student' or 'Teacher'.

3.4 Student 360° Profile & Portfolio Inspection Modal

Teachers and administrators can click **'View Student Profile'** on any enrolled student card to inspect institutional data (UDISE code, school, district, guardian), real-time course completion percentages, quiz score histories, and submitted handwritten answer sheets.

4. Complete Postman API Collection Catalog

A full Postman Collection (v2.1.0) has been generated and mounted at /static/docs/Nexus_LMS_Postman_Collection.json. You can import this JSON directly into Postman, Bruno, or Insomnia by URL or file download.

Category / Module	HTTP Method & Path	Description & Governance Rules
Authentication	POST /login GET /logout GET /profile POST /profile/update	User authentication (Admin, Teacher, Student). Returns JWT cookie. Profile displays read-only assigned departments for teachers.
Faculty Governance	POST /department/teachers/assign-departments GET /api/teachers/{id}/assignable-courses POST /department/teachers/assign-course	Enforces Step 1 (Dept assignment) → Step 2 (Assignable course filtering) → Step 3 (Course assignment). Rejects mismatched departments.
Ans Sheet Upload	POST /api/course/{id}/upload-answer-sheet GET /api/course/{id}/answer-sheets POST /api/answer-sheet/{id}/evaluate GET /department?tab=answer-sheets	Student PDF answer upload, course submission list, teacher evaluation & grading, and Admin surveillance KPI window.
Student 360°	GET /api/students/{id}/profile-modal	Returns student institutional data, enrolled courses, completion rates, quiz attempts, and submitted answer sheets.
Live Classes	POST /api/live-classes/create POST /api/live-classes/{id}/status POST /api/live-classes/{id}/questions POST /api/live-classes/questions/{id}/answer	Live streaming class scheduling, Go-Live status toggle, student live questions, and faculty question resolution.
1-on-1 Mentorship	POST /api/mentoring/student-request GET /api/mentoring/my-sessions POST /api/mentoring/{id}/complete	Student faculty appointment requests, scheduled slot confirmation, Google Meet links, and session outcome logging.
Mylo AI (Gemini)	POST /teacher/course/{id}/quiz/ai-generate POST /api/mylo/test GET /api/mylo/usage-metrics	Google Gemini dynamic quiz generation, Bloom's taxonomy prompts, connection validation, and latency analytics.
PowerBI APIs	GET /api/v1/reporting/students GET /api/v1/reporting/courses GET /api/v1/reporting/quiz-attempts GET /api/v1/reporting/engagement	High-throughput JSON feeds for enterprise analytics dashboards (PowerBI, Tableau, Looker) with student portfolios.

5. Postman Collection Import Instructions

Option A: One-Click URL Import in Postman

1. Open Postman → Click **Import** (top left).
2. Paste the live raw collection URL:
https://boats-revisions-released-strap.trycloudflare.com/static/docs/Nexus_LMS_Postman_Collection.json
3. Click **Import**. The entire collection with 35+ organized requests will load immediately.

Option B: Local Download & Drag-and-Drop

Download the JSON file directly from http://127.0.0.1:8000/static/docs/Nexus_LMS_Postman_Collection.json and drag it into Postman.